CNC

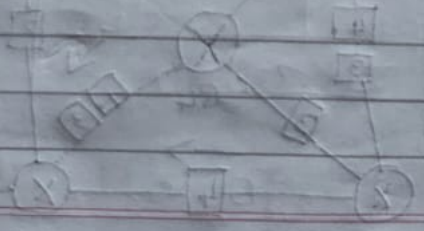
* Routing Algorithm

1) Dijikstra's algorithm (formula)
2) Flooding
3) Distance Vector Routing (problem) (4-5 points)
4) Count-to-Infinity Problem. (exp with diag)
5) Link State Routing (steps) - explain 25,28
6) Hierarchial Routing (simple diag) - explanation.
7) & Traffic shifting
8) Leaky and Token Buckets (exp & diff) (diag)
9) Fair Queuing & weighted Fair Queuing.
10) Differentiated Services.

MSE II

Q) Service provided at the network layer

- packetizing
  - The network layer divides data into packets for transmission across the network
- Routing :
  - It finds the best path for data to travel from source to destination
- Error Control :
  - Detects and helps correct errors that occur in the transmitted packets
- Flow control :
  - Ensure that data is sent at a rate that the receiver can handle
- Congestion Control
  - Prevents network overload by controlling the amount of data entering the network
- Security
  - provides protection of data from unauthorized access and attacks during transmission

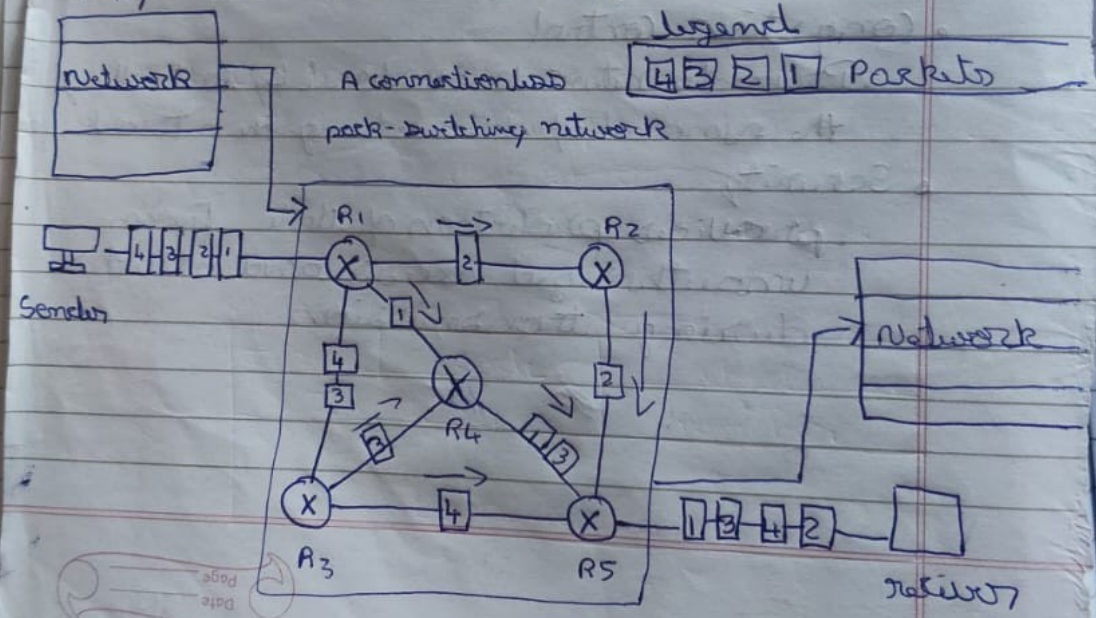


Packet Switching

→ • In packet switching, data is divided into small packets before sending
• Each packet travels independently through the network.
• Used in computer networks like the internet

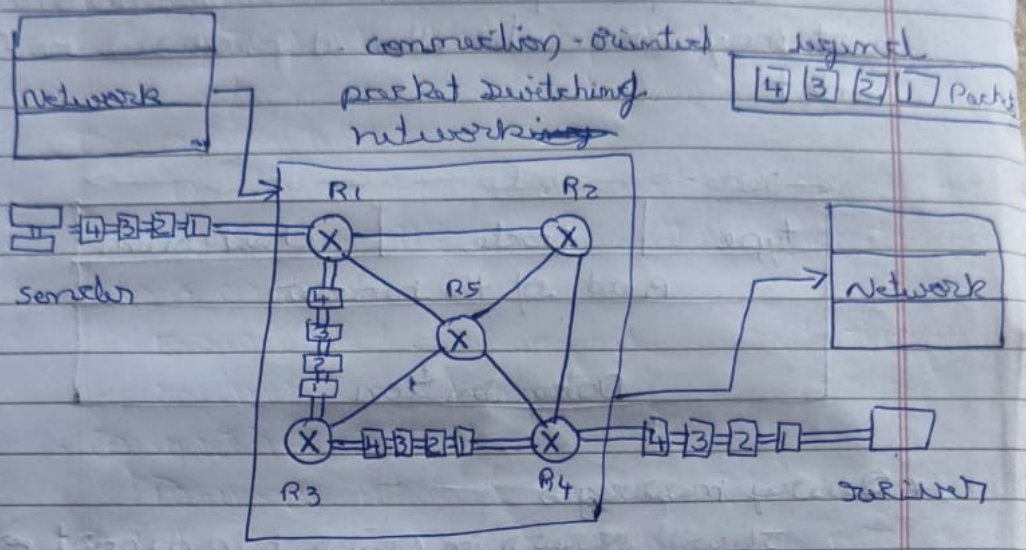
modes of packet switching

1) Datagram Approach
* Each packet is treated independently
* Packets may take different routes to reach the destination
* Packet can arrive out of order and are reassembled at the receiver end.
* Eg: Internet (IP network).



ii] Virtual Circuit Mode:

- A logical path is set up before sending packets
- All packets follow the same route
- Packets arrives in order
- Eg: Frame Relay, ATM network



3] Notations

i] Binary notations (base 2)

ii] dotted-decimal notation (base 256)

iii] hexadecimal notation (base 16)

Binary 10000000 00001011 00000011 00011111

Dotted decimal 128.11.3.31

hexadecimal 800B031F

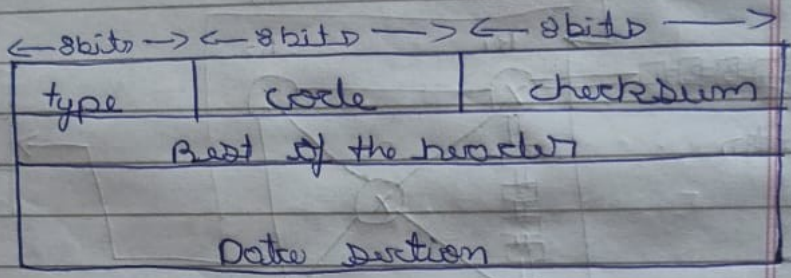
4] Internet control message protocol version 4 (ICMPV4)

-> • used in IPV4 for error reporting and network testing & also find and fix communication problems

• ICMP message are divided into two broad categories

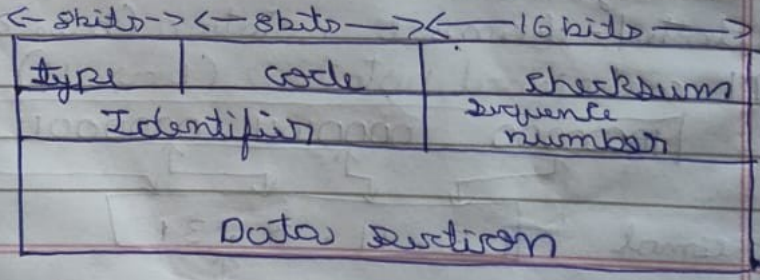
* Error reporting message

It report problem that a router or a host may encounter when it processed an IP packet



* Query messages

It will occurs in pairs, help a host or a network manager get specific information from a router or another host.



5 Comparision of IPv4 & IPv6

IPv4	IPv6
* 32 bits	* 128 bits
* Decimal, separated by dots (.)	* hexadecimal, separated by colons (:)
* Around 4.3 billion (2^{32}) addresses	* Approximately 3.4×10^{38} (2^{128}) addresses
* 20-60 bytes of header	* fixed 40 bytes header
* fragmentation done by sender and routers	* fragmentation done only by sender
* less efficient due to large routing table	* more efficient with hierarchical addressing
* It is older & runs out of addresses	* It is newer, faster & more future with a huge number of unique addresses
* header includes a checksum field	* does not have a checksum field
* Supports broadcasting	* does not support broadcast, uses multicast.

6 The delays in network can be classified into 4 types

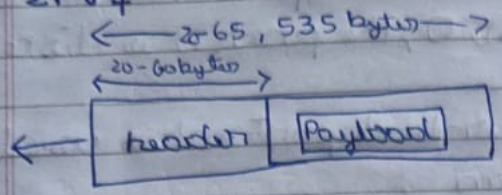
i) Transmission Delay:
* Time taken to push all bits of a packet onto the link
* Depends on packet size and transmission rate
* $Delay_{tr}$ = Packet size / Transmission rate

ii) Propagation Delay
* Time taken for a bit to travel from sender to receiver
* Depends on distance and signal speed
* $Delay_{Pg}$ = distance / propagation speed

iii) Processing Delay:
* Time routers or hosts take to check for bit error and decide where to send the packet next

iv) Queuing Delay:
* Time a packet waits in a queue before being transmitted
* Depends on network traffic load.

7 IPv4



IP datagram

0	4	8	16	31
VER 4 bits	HLEN 4 bits	Service type 8 bits	Total length 16 bits	
Identification 16 bits			Flags 3 bits	Fragmentation offset 13 bits
Time to live 8-bits		Protocol 8-bits	Header checksum 16 bits	
Source IP address (32 bits)				
destination IP address (32 bits)				
options + Padding (0 to 40 bytes)				

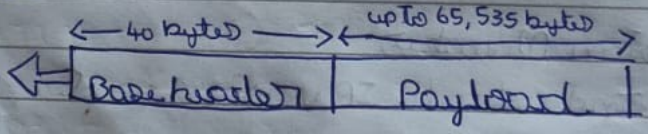
header format

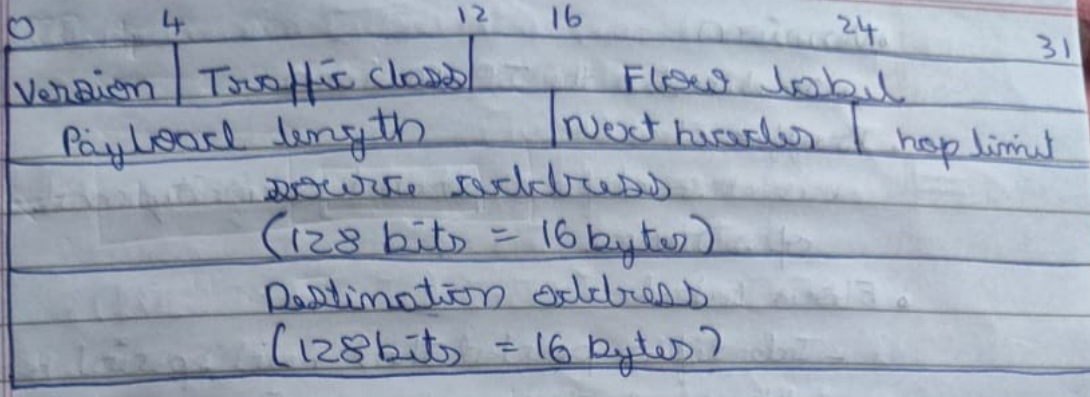
- Version Number
  - * Indicates the IP version used
- header length
  - * Stores the length of the IP header in 32-bit words
- Total Length
  - * Total size of the datagram (header + data) in bytes
- Time to Live
  - * Limits how long a packet can stay in the network

ternet Control ...

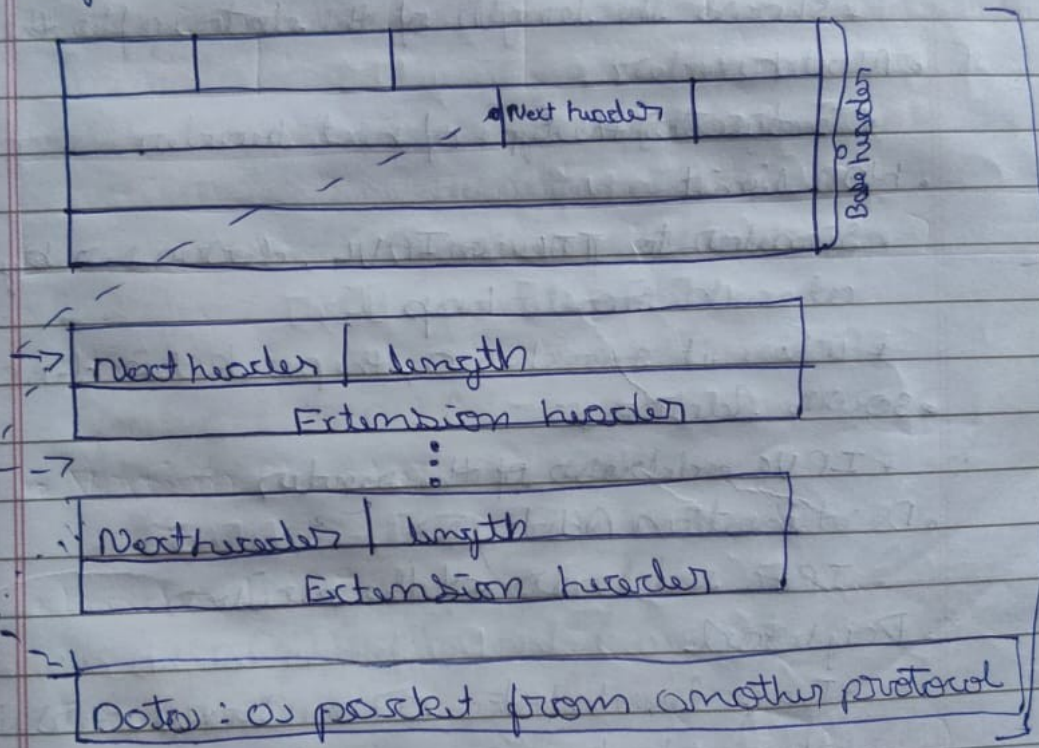
decreased by 1 at each hop.
* when it reaches 0, the packet is discarded
- Protocol
  * Tells which higher-level protocol is used
- header checksum
  * used to detect errors in the IP header
- Source Address
  * IP address of the sender
- Destination Address:
  * IP address of the receiver
- options:
  * used for extra features like security or route recording
- Payload
  * The actual data carried by the packet

8 IPv6





Payload in an IPV6 diagram



- Version
  - Indicates the IP version
- Traffic class
  - used to define packet priority and quality of service
- Flow label
  - Identifies packet that need special handling like real-time streaming
- Payload length
  - Shows the length of the data after the header
- Next header
  - Indicates the type of next header
- HOP limit
  - Similar to TTL in IPV4, decreased by 1 at each router hop
  - When it reaches 0, the packet is discarded
- Source Address:
  - IPV6 address of the sender
- Destination Address
  - IPv6 address of the receiver
- Payload
  - The actual data or message carried by the packet.



9 Unicast Routing

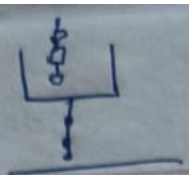
- ~~Definition~~ The packets are sent from one source to one specific destination
- The packet is routed hop by hop from the source router to the destination router using forwarding table
- Source host doesn't need forwarding table
- Destination host also does not need a forwarding table
- Only routers that connect network together maintain forwarding tables
- It'll deliver the packet efficiently from the source router to destination router.

10 Dijkstra's algorithm() {

```
Tree = {root}
for(y=1 to N){
  if (y is the root)
    D[y] = 0
  else if (y is neighbor)
    D[y] = c[root][y]
  else
    D[y] = ∞
}
repeat{
```

find a node w, with D[w] minimum among all nodes not in the Tree.
Tree = Tree ∪ {w}
for (every node x, which is neighbor of w and not in the Tree) {
D[x] = min{D[x], (D[w] + c[w][x])}
}
} until (all nodes included in the tree)
}

// Distance vector Routing () {
D[myself] = 0
for (y = 1 to N {
if (y is a neighbor)
D[y] = c[myself][y]
else
D[y] = ∞
}
Send vector {D[1], D[2], ---, D[N]} to all neighbors
repeat (forever) {
wait (for a vector Dw from a neighbor w or any change in the link)
for (y = 1 to N) {
D[y] = min [D[y], (c[myself][w] + Dw[y])]
}



13 Token Bucket Algorithm

It is simple and rate limiting mechanism

- Token Bucket
  - * A bucket holds tokens and each tokens sends a fixed amount of data.
  - * The bucket has maximum capacity, the extra tokens are discarded if it becomes full
  - * Tokens represent permission to send data
- Request Processing
  - * When packets arrives
    - if enough tokens are available → the packet is transmitted immediately
    - if not enough tokens → the packet is delayed or dropped
  - * This ensure that the packet are sent at a controlled rate
- Token Refill
  - * Tokens are added to the bucket at the constant rate
  - * This allows occasional bursts of data, but keeps the average rate limited over time
- Key characteristics
  - * Simplicity
  - * Smooth rate limiting • Burstiness

14 Leaky Bucket Algorithm

-> It is a rate limiting mechanism that enforces a constant output rate by continuously leaking excess traffic or requests.

- Leaky Bucket
  - * used for traffic shaping in network
  - * acts like bucket with small hole
  - * Packets arrives at varying rates but leaves at a fixed rate
  - * If the packet is full extra packets are dropped.
- Request Processing
  - Incoming packets are stored in the bucket
  - * Packets leak out one by one at a constant rate
  - * If the bucket overflows, extra packets are discarded.
- Key characteristics
  - * Constant output, smoothing

